NEOSTATS DATA SCIENCE HACKATHON · LOAN REFUSAL PREDICTION

Predicting whether a loan application will be refused — at the moment it arrives

1.33M
decided applications
(Approved + Refused)

21.9%
refusal rate
≈ 1 refusal per 3.6 approvals

37%
refusal rate in the most
recent 12 months (OOT)

0.904
test ROC-AUC of the
final LightGBM model

Business problem

Every refused application still costs underwriting time. Predicting refusal at the point of application lets the lender pre-screen likely refusals, fast-track safe ones, and steer customers to an amount or product they can get.

Target: REFUSED

Built from `NAME_CONTRACT_STATUS`: Approved → 0, Refused → 1. **Canceled** and **Unused offer** removed — there the *customer*, not the lender, decided. Moderate imbalance (21.9% positives): accuracy is misleading — "always approve" scores 78%.

1 Clean & join
2 Remove leakage
3 Point-in-time features
4 Time-based split + OOT
5 LogReg vs LightGBM
6 Threshold & risk bands
7 Power BI dashboard

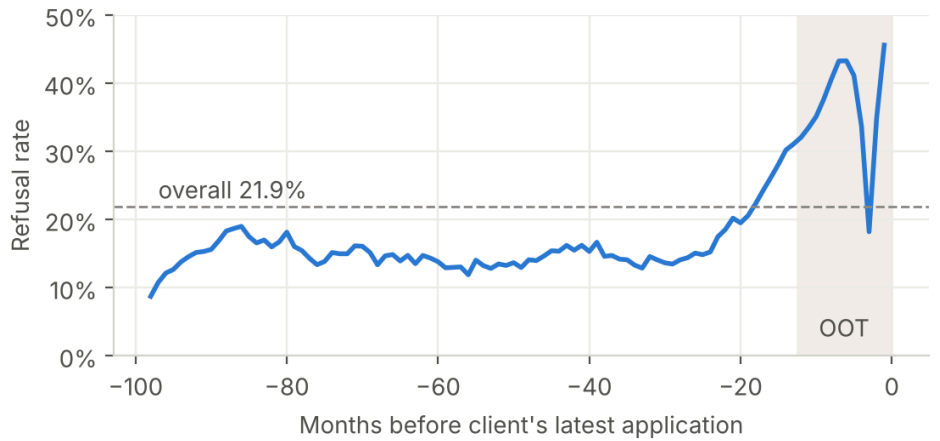
Data: previous_application (main) + client attributes on SK_ID_CURR

DATA PREPARATION & VALIDATION DESIGN

Leakage removed, features built point-in-time, and a strict out-of-time test

- **Join:** applications → clients on `SK_ID_CURR`, validated many-to-one; row count unchanged, 100% matched (train + test client files).
- **13 leakage columns removed** with evidence — e.g. payout & repayment dates and insurance-on-approval are missing for 100% of refusals; `CODE_REJECT_REASON` is the answer itself; `AMT_CREDIT` is the *granted* amount.
- **Hidden time leak avoided:** client tables describe the *latest* application, so age and job tenure are recomputed at each decision date; latest-loan columns are never used.
- **Odd values:** 365243 → missing (+ flag), XNA → missing, XAP → "Not applicable".
- **15 engineered features** — amount ÷ income, financed share, age/income/employment bands, weekend & hour bands, external-score mean & missing count, and **history from strictly earlier applications** (count, refusals, days since last).
- **Imputation & encoding fitted on train only**, inside sklearn pipelines.

SPLIT	RULE	ROWS	REFUSAL
Train 70%		714,467	17.3%
Validation 15%	DAYS_DECISION ≤ −365, stratified	153,100	17.3%
Test 15%		153,101	17.3%
OOT	last 12 months	306,791	**37.3%**

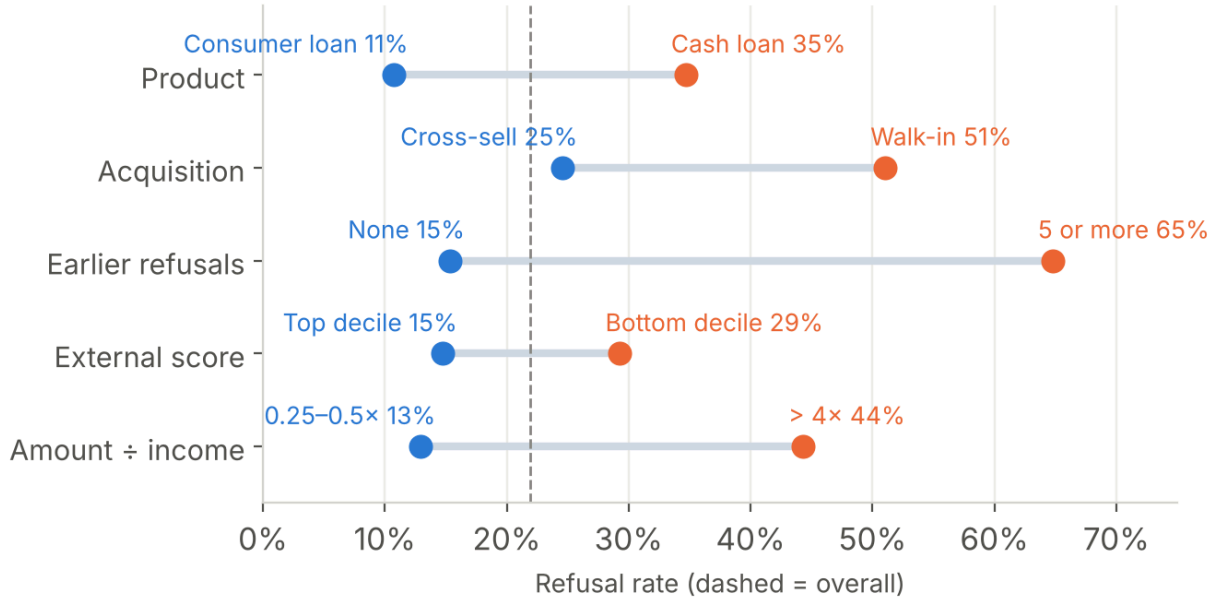


The refusal rate more than doubles in the OOT window — a drift the model must face.

EXPLORATORY ANALYSIS

Product, credit history and affordability drive refusals — demographics much less

1. **Cash loans are refused 34.7%** vs 10.7% for consumer (POS) loans — 3.2×. Fast-track POS; review cash.
2. **Walk-ins: 51.0%** refused vs 24.6% for cross-sell offers (2.1×) — pre-screened customers convert better.
3. **History repeats:** 15.4% refused with no earlier refusal, **64.8%** after 5+ (4.2×). Guide repeat applicants to a different amount/product.
4. **Affordability:** asking > 4× annual income → 44.3% refused vs 13.0% at 0.25–0.5×.
5. **Drift:** last 12 months 37.3% vs 17.3% before (2.2×), with a shift towards cash loans.
6. **Surprise:** repeat clients are refused 28.4% vs 4.9% for new clients — even within cash loans. Income and age bands differ by only 6 and 2 points.



Associations, not causes — e.g. product mix partly explains the client-type gap.

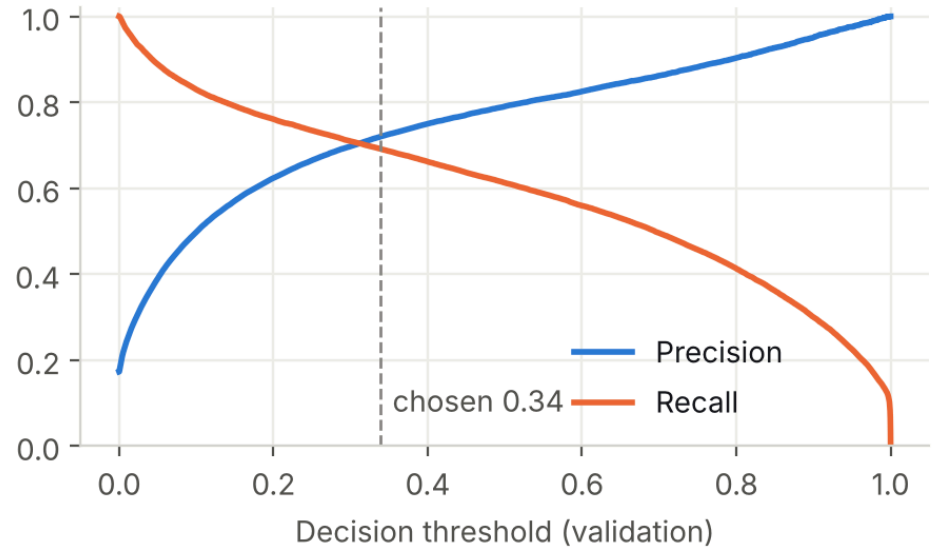
MODEL COMPARISON (VALIDATION SET)

LightGBM beats the logistic baseline by +0.12 PR-AUC; unweighted version chosen for honest probabilities

MODEL	ROC-AUC	PR-AUC	PRECISION	RECALL	F1
Logistic Regression (class-weighted)	0.855	0.652	0.589	0.586	0.588
LightGBM, class-weighted (scale_pos_weight 4.8)	0.906	0.773	0.720	0.696	0.708
LightGBM, unweighted + tuned threshold ✔	**0.906**	**0.774**	**0.721**	**0.691**	**0.706**

- **Why LightGBM:** boosted trees capture non-linear effects and interactions (cash loan × earlier refusals), handle missing values and categories natively. Tree size (31/63/127 leaves) chosen on validation; early stopping on validation PR-AUC.
- **Imbalance:** class weights vs. no weights + threshold tuning give the same ranking; the unweighted model has better-calibrated probabilities (Brier 0.074 vs 0.083) — important for risk bands.
- **Main metric = PR-AUC** (focus on the minority class); accuracy is not used.
- **Reliance check:** dropping history features costs 0.07 PR-AUC; no single column carries the model → no hidden leak.

Threshold = 0.34 (max F1 on validation)



A flag routes the application to **faster pre-screening**, not automatic rejection. A false alarm costs a short review; a missed refusal wastes a full underwriting cycle — so we balance both (precision 0.72, recall 0.69). The threshold can be moved for the business.

TEST VS OUT-OF-TIME

Stable on test; on newer data the ranking holds but refusals are under-predicted

FINAL MODEL	ROC-AUC	PR-AUC	PRECISION	RECALL	F1
Validation	0.906	0.774	0.721	0.691	0.706
Test (scored once)	0.904	0.772	0.716	0.688	0.702
OOT — last 12 months	0.846	0.806	0.806	0.563	0.663

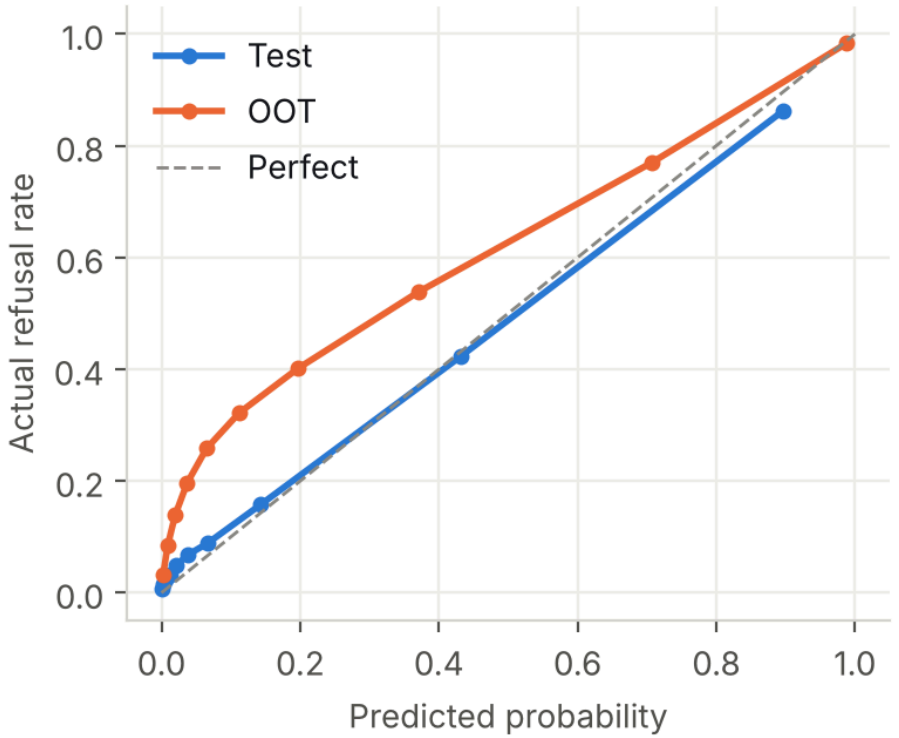
- **Test ≈ validation** → generalises to unseen applications; threshold transfers.
- **OOT:** ROC-AUC 0.904 → 0.846, recall 0.69 → 0.56. Refusals jumped to 37.3% but only 26.0% are flagged; score PSI 0.22 (moderate shift). PR-AUC rises only because positives are more common.
- **Action:** retrain / recalibrate on recent months, lower the threshold, monitor PSI monthly.

Risk bands (cut-offs set on validation)

BAND	SHARE (TEST)	REFUSAL TEST	REFUSAL OOT
Low	77.4%	5.0%	18.0%
Medium	6.0%	24.8%	44.0%
High	16.6%	71.6%	80.6%

Low < 0.17 (≤ 5% refusals → fast-track) · Medium 0.17–0.34 · High ≥ 0.34 (pre-screen)

Calibration

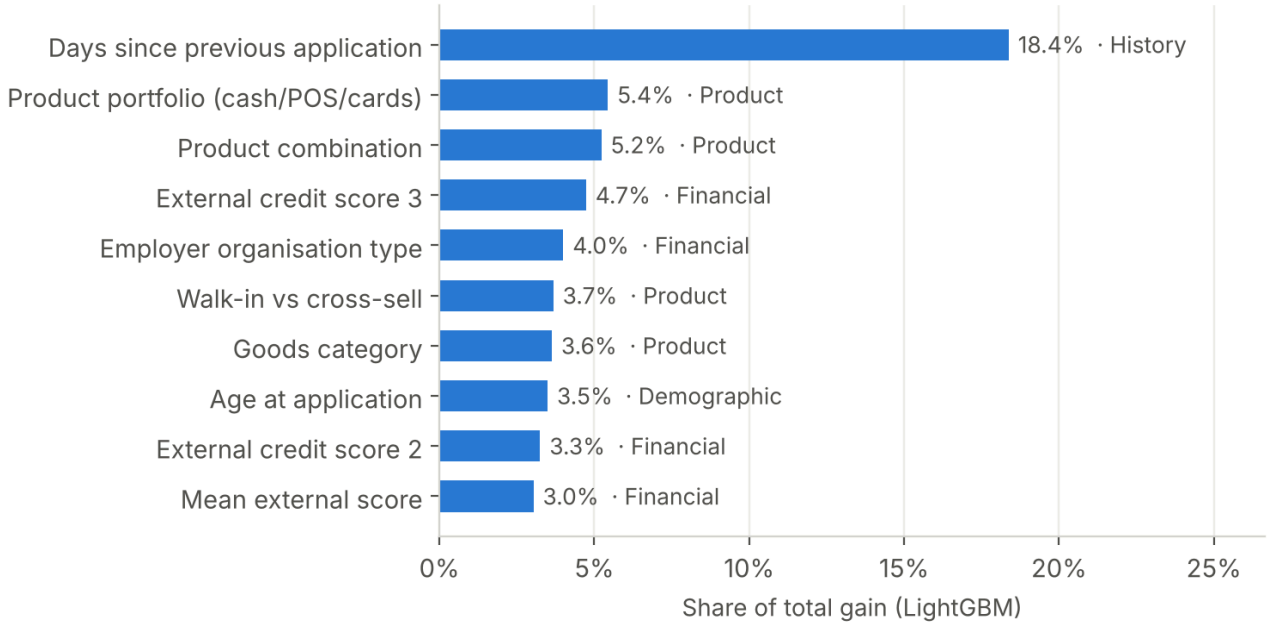


Test sits on the diagonal; OOT above it → the model learned the older 17% world.

Bands rank correctly on both sets: Low < Medium < High

GLOBAL FEATURE IMPORTANCE — FINAL LIGHTGBM

Re-application timing, product and external scores drive the predictions



By feature group

Product	33%
Financial	28%
History	26%
Demographic	8%
Channel	5%

- **Re-applying soon after a previous application** is the strongest signal — typical after an earlier refusal.
- **Product** (portfolio, combination, walk-in vs cross-sell) and **external scores** follow.
- Gender is **not** a model feature (fairness); demographics carry only 8%.

Caveat: gain importance shows predictive usefulness, not cause and effect or direction; correlated features share credit (portfolio ranks #2, yet dropping it costs nothing).

Exported to outputs/feature_importance.csv for the dashboard